# CLASSIFICATION OF ECG SIGNAL USING MACHINE LEARNING TECHNIQUES

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*Abstract*— **Electrocardiogram (ECG) signals are the impulses generated by the heart which are used to analyze the proper functioning of heart. Our work deals with the efficient analysis of Electrocardiogram (ECG) signals imported from MIT-BIH database into MATLAB platform, generation of the imported ECG signal, pre-processing the generated signal to remove the noises mainly the baseline wandering and power line interference from which features are extracted. For adequate study of the ECG signal Daubechies and Haar wavelet techniques are compared. Proper decomposition of the signal is achieved using Db4 and Db5 Daubechies wavelets as their scaling functions are analogous to ECG signal. PAN TOMPKINSONS algorithm is considered in our study as it serves best for precise identification of most prominent features namely QRS complexes, RR interval's as they constitute the major data required for clinical analysis and research. After feature extraction ECG signals are trained using machine learning techniques for detecting the presence of Arrhythmia using different classifiers adopting Weka software.**

**Keywords— ECG, Notch filter, Daubechies wavelets, Haar wavelet, feature extraction, SVM classifier, Neural Networks, MLPNN**

## 1 INTRODUCTION

Nowadays cardiac malfunctions are quite recurrent; hence a proper diagnosis is an essential concern. Electrocardiogram (ECG) signals are the impulses generated by the heart which are used to analyze its behaviour. Each beat consists of PQRST waves generated by contraction and expansion of heart muscles (atria and ventricles). A proper monitoring of these beats can help us detect the abnormalities i.e., arrhythmia in the heart. Hence an early diagnosis of arrhythmias can prevent heart strokes.

Cardiovascular diseases (CVD's) are a chronic disease which claims more lives than all forms of cancer and other chronic disorders. In India, quarter of all mortality is due to CVD's. As per the clinical study, average of about 1 out of 4 dies every 38 seconds because of this disease. A medical practitioner usually approves for ECG for those who have risk of heart diseases due to their family background or their irregular habits which leads to obesity, high blood pressure, etc, hence a proper diagnosis of the rhythmic behavior of heart using ECG waveforms is indispensable to detect asymmetry present such as heart beats, strokes, and enlargement of the heart, blockages and many more. Accurate classification of the ECG signals based on the features extracted plays a vital role in the diagnosis of CVD's. This study mainly focuses on classification of ECG signal using wavelet transforms. It includes ECG signal generation from MIT-BIH arrhythmia dataset, pre-processing (noise reduction and baseline adjustment), feature extraction and finally classification of signal to detect arrhythmia using artificial neural networks (ANN) techniques.[1]

### I. 1.1 BACKGROUND KNOWLEDGE

Electrocardiography is the process of producing electrocardiograph signals (ECG or EKG) which is the graph plotted between voltage versus time produced due to the electrical activity of the heart. The ECG signal basically consists of several beats and each beat comprises of P wave, QRS complex, T wave and U wave respectively. Each peak (P, Q, R, S, T, and U), intervals (PR, RR, QRS, ST, and QT) and segments (PR and ST) of ECG signals have their corresponding amplitude values. These peaks, intervals, and segments are called ECG features. These waves are associated with Atrial and ventricular repolarisation and depolarisation. Anomalies in the rhythmic function of the heart results in Arrhythmia. Fig 1 depicts these features for one ECG cardiac cycle.

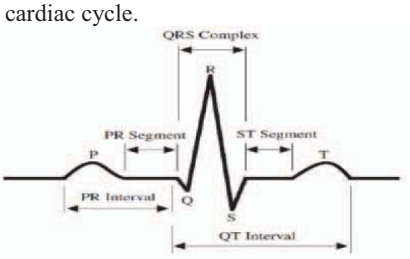


Fig1

The amplitude and timing information of a normal ECG signal can be inferred through tables1 and 2

| Wave | Amplitude |
|---|---|
| P wave | 0.25 Mv |
| R wave | 1.6 Mv |
| Q wave | 25% of R wave |
| T wave | 0.1 to 0.5 mV |

Table1

| Interval | Duration |
|---|---|
| PR wave | 0.12-0.2 sec |
| QT wave | 0.35-0.44 sec |
| ST wave | 0.05-0.15 sec |
| P wave interval | 0.11 sec |

Table 2

ECG signal analysis, classifying plays a major role in clinical analysis of a signal which further helps in depicting the type of abnormality present in the heart signal. Major concern in diagnosing the type of abnormality is that ECG signal will differ from each individual as heart rate of individual may alter based on his psychological and mental conditions. Lack of standardization of ECG signal is the major challenge involved in classifying the ECG signal.

Various steps involved in classifying ECG signal are pre-processing the signal, feature extraction and normalization of the signal. Different types of noises occur during acquisition of the signal. Pre-processing involves the removal of noises present in ECG signal. Noises due to Power line interference, Base line wandering, Electromyogram, motion artefacts need to be mitigated using efficient filters. Feature extraction includes finding the peaks and intervals in ECG signal. Based on the features extracted patient's heart functionality is properly diagnosed for the detection of arrythmia using machine learning algorithms.

### 1.2 NEED FOR CLASSIFICATION

Classification of ECG signal can help in identifying the abnormality present in ECG signal of a patient. After identifying the abnormality, the heart diseases can be detected and the better treatment of the patient can be done. Classification of ECG signals is a challenging problem. There are many issues involved in classification process of the ECG signal. Major issues in ECG classification are lack of standardization of ECG features, variability amongst the ECG features, variations in the beat in single ECG, individuality of the ECG patterns, non-existence of optimal classification rules for ECG classification and variability in ECG waveforms of patients. We can be able to diagnose a new patient more accurately than manually after precise classification which helps in providing better treatment to the patients.

### 1.3 PROPOSED METHODOLOGY FOR CLASSIFICATION

Classification of ECG signal involves steps like pre-processing, feature extraction and finally using machine learning algorithms for accurate detection of arrhythmia. Pre-processing the ECG signal involves removal of different kinds of noises which can be removed using various kinds of filters like linear phase high pass filter, low pass linear phase filter, notch filter, median filter, mean median filter etc.,[2]

Feature extraction involves detecting the peaks of the signal. Feature extraction techniques like Pan Tompkins algorithm, continuous wavelet transform (CWT), Discrete Wavelet Transform (DWT), Discrete cosine transform (DCT), S Transform (ST), Continuous Wavelet Fourier transform (DFT), Principal Component Analysis (PCA), Daubechies wavelet (Db4), Haar wavelet.[4]

Thus, classification involves the usage of machine learning algorithms like multilayer-perceptron neural network (MLPNN), Support vector machine (SVM), Quantum Neural network (QNN), Probabilistic neural network (PNN), Type2 Fuzzy Clustering Neural Network (T2FCNN), etc. based on the features extracted, adopting WEKA software.

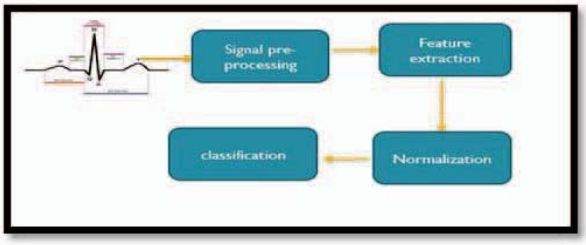


Fig 2 Block Diagram

## 2. ECG DATA PROCESSING

### 2.1 ECG Database Collection

The MIT-BIH arrhythmia data-base is the standard material used for training & testing the algorithm developed for detection & classification of arrhythmia ECG signals. There are total 48 records. All signals are two lead signals denoted as lead A & lead B signal. Lead represents the position of measuring the heart signal in each individual. Out of 48 records 23 records are normal sinus rhythm (NSR) and remaining 25records show clinical cardiac abnormalities. All these 48 recordings are taken for our study. Each recording is the information of single individual which helps us to know the presence of Arrhythmia, causes of arrhythmia, abnormalities, medication given to that particular patient. There are certain recordings where there is no abnormality and no Arrhythmia. These 48 signals are used for testing and training the signal. All these data sets are collected from Physio-net bank. Physio-net bank contains different kinds of datasets which includes EMG signals, ECG signals, BIDMC signals etc., which helps researchers to continue their study regarding different diseases. [7]

In our study we considered the ECG signal of a patient who has typical Arrhythmia condition. This is the recording of a male patient aged 60. The considered position for analysis is MLII lead. It gives information regarding PVC, APC. PVC is premature ventricular contraction which is a medical condition where ventricles in our heart are contraction before the expected time that leads to improper function of heart. APC is premature articular contraction; this is the phenomenon where articles are contracted too early. As per the clinical study it is found that PVC shows more impact on Arrhythmia than APC. The patient has supraventricular entropy where there is abnormal heart beat due to improper electrical activity in upper part of the heart.

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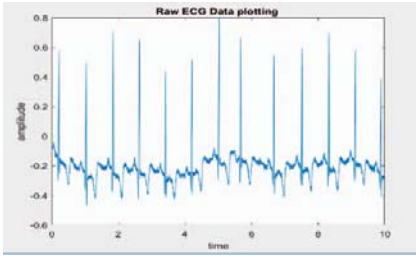


Fig. 3(a) ECG signal of record No. 100

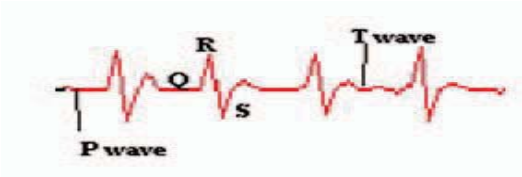


Fig. 3(b) Normal beats

2.2 Pre-processing ECG signal

The ECG signal has to be pre-processed to remove the noises present in the high, medium and low frequency ranges. Noises like Electromyogram noise, power line interference, Base line noises are major types of noises which are to be pre-processed before extracting its features. These noises are mainly due to environmental and biological interferences.

Electromyogram noise (EMG noise) occur due to biological orelse technical sources indicating the ratio between the muscle contraction and unwanted noises.

The environmental interference is mainly due to improper grounding, wire coiling, etc which cause power line interference mainly at 50 Hz which can be removed by using IIR filters.[2]

Biological interferences are due to patient's respiration, any motion of the patient, etc which are mainly present in the low frequency ranges. These cause baseline wandering which can be rectified using wavelet transforms,

In our study, we analyzed and reduced noises caused due to PLI and base-line wandering using different filters. We have incorporated Butterworth Notch filter for reducing Power line interference of the signal as it has high attenuation. Detrending and smoothening of the signal is done using wavelet technique. Daubechies wavelets are used for Baseline removal in the ECG signal. Below is the flowchart indicating the steps involved in the signal pre-processing.

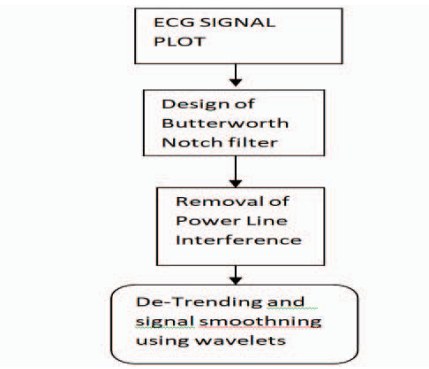


Fig 4 Flowchart for signal pre-processing

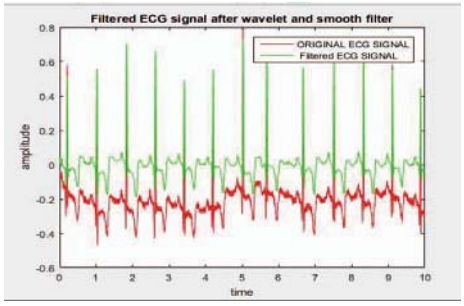


Fig 5 Signal after Pre-processing

Fig 5 is the pre-processed signal after the removal of noises due to power line interference and baseline wandering, x axis indicates the time and y- axis denotes the amplitude of ECG signal. The figure 5 compares the original signal (red) with the pre-processed signal(green).

2.3 Wavelet Transform

Wavelet transform plays a significant role in the analysis of ECG signal. There are various kinds of wavelet transforms like Haar, Daubechies, Coiflet, Hilbert, etc and its selection purely depends on our application. Wavelet analysis help in the removal of noises and improve the signal strength. In our work, we dealt with the decomposition the ECG signal into 8 level using Haar and Daubechies wavelet techniques and thus performed a comparative analysis of both the wavelet transforms which would thereby lead to accurate extraction of features. For a detailed analysis of wavelet transform refer [3].

2.4 Haar Transform

Haar transform is the simplest of all waveforms. This wavelet family consists of square integrable functions on a scale of [0,1], due to which its technically discontinuous therefore not differentiable. But this property of the Haar wavelet transform helps it to be widely in the application of analysis of signals with sudden transitions, for example monitoring of machine failures.
Haar mother wavelet function is defined as $\psi(t)$

$$\psi(t) = \begin{cases} 1, 0 \le t < 1/2 \\ -1, 1/2 \le t < 1 \\ 0, otherwise \end{cases}$$

This transform cross multiplies a function for example sine function against Haar function $H_n$, where n denotes the order of the matrix. For our study we cross multiplied an $H_8$ matrix with the extracted ECG signal.

Thus, the Haar transform is given by:

$$y_n = H_n x_n$$

Where $y_n$=transformed function

$H_n$=Haar matrix

$x_n$= Input signal

Below is the 8-level decomposition of ECG signal using Haar wavelet transform.

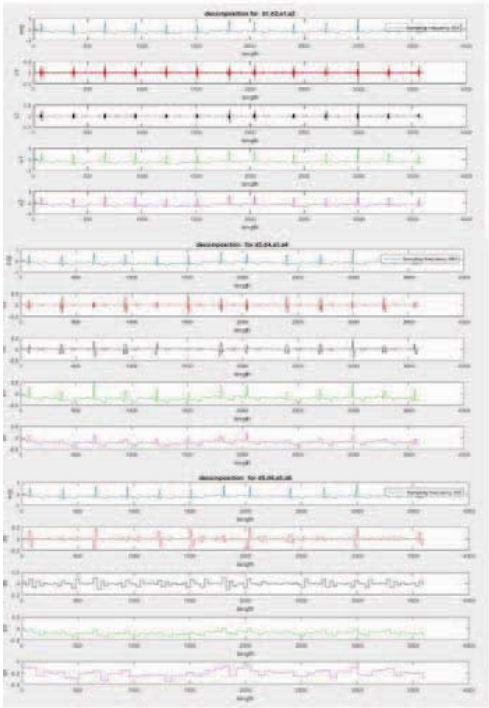

Fig 6 Eight level decomposition using Haar wavelet

The above figure clearly shows us that Haar wavelet is not apt for ECG signal analysis. This wavelet transform is not continuous and this is found to be more suitable for square waveforms which doesn't resemble ECG signal.

2.5 Daubechies Transform

This is another member of the wavelet family defining discrete wavelet transforms which is quite complex but is widely used due to its property of retrieval of lost data after decomposition thereby improving accuracy of the results.

These wavelets are chosen to have the highest number of vanishing moments A, that doesn't imply smoothened function based on the support width given by 2A-1.this wavelet is has two naming schemes, DN where N denotes the number of taps and dbA which refers to the number of vanishing moments, after the application of this wavelet we get $2^{A-1}$ possible solution from which the desired waveform is chosen for further analysis. In special cases Haar wavelet is similar to db1 wavelet. The vanishing moment that is A limits the signal's ability of polynomial behavior of the signal. For example, D4 encodes signal with 2 polynomial coefficients which infers to one constant component and linear signal component.

In our study of signal classification, we choose db6 Daubechies wavelet series due to similarity to the ECG signal. db6 wavelet are biorthogonal, asymmetric and orthogonal wavelet transforms. This wavelet encodes 3 polynomial coefficients i.e. constant, linear and quadratic signal components.

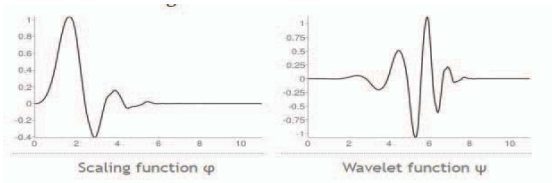


Fig 7(a) db6 Wavelet Transform

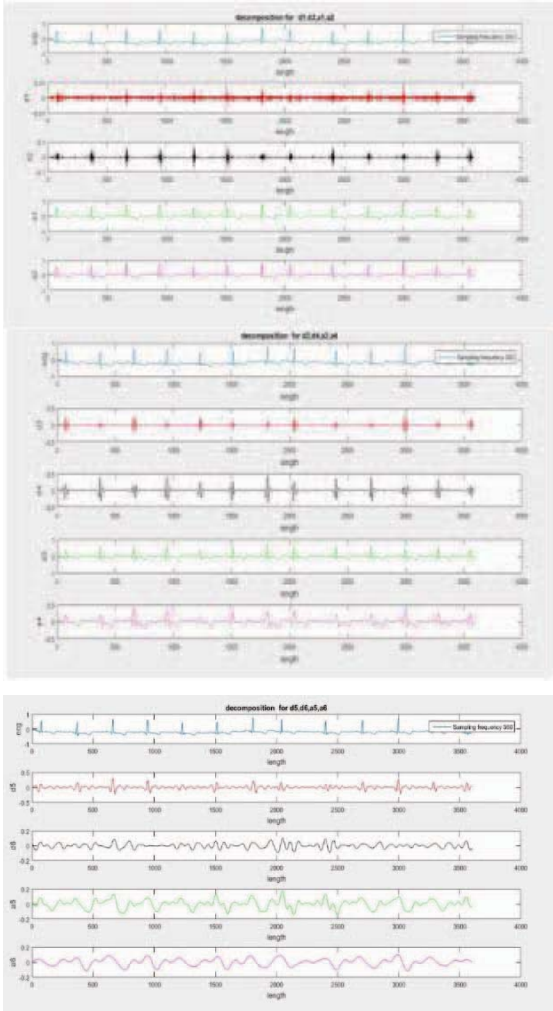

Fig 7(b) Signal decomposition after applying db6 wavelet

The above fig 7(b) illustrates about the ECG signal decomposition using Daubechies wavelet series, thus we get to know that a4 and a3 signal show similarity to the ECG signal.

## 3. FEATURE EXTRACTION

### 3.1 DETECTION OF R PEAK & QRS COMPLEX

After decomposition of the signal features are extracted using Pan Tompkins algorithm. QRS complex is effectively detected using this algorithm. Preprocessing, derivating, squaring, averaging, thresholding, searching back for missed QRS complexes, elimination of multiple QRS complexes are the various steps involved in this algorithm.

Preprocessing is done using Band pass filter, high pass filter in order to remove major noises involved in signal. More accurate QRS complex can be found after derivating the signal in order to get the information regarding the slope of the signal. After derivating the signal is squared in order to make all the information from the data come into positive axis which further helps in easy detection of peaks. After squaring the signal is averaged to remove the high frequency noise in the signal. Decision rules like Fiducial marks need to be taken into consideration. Fiducial mark can be determined for the desired waveform as per the first rising edge of the signal. Time duration of the rising edge should be equal to the width of the signal. Adjusting the thresholds is the next step which needs to be performed. Two threshold values are set accordingly. Higher of the two thresholds is considered for first analysis of the signal. If there is no QRS complex detected in a certain time period the lower threshold is used and the search back technique is utilized to look back in time for the QRS complex. Values of each threshold and R-R interval limits are set automatically by the developed algorithm. Detailed description along with the implementation of Pan-Tompkins algorithm is discussed in paper [12][5].

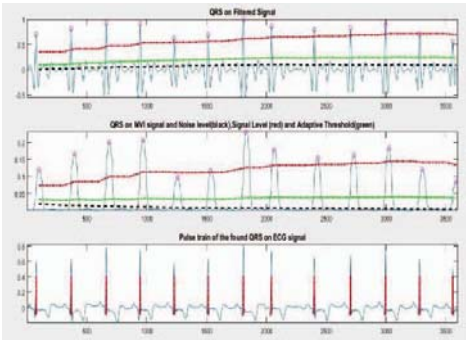


Fig 8(a) QRS complexes of record 202

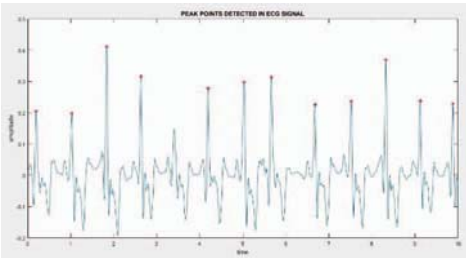


Fig 8(b) R-peaks of record 202

Figure 8(a) determines the QRS complexes, its strength and the noises levels in the signal whereas figure 8(b) indicates the R-peaks.

## 4. CLASSIFICATION OF ECG SIGNAL

### 4.1 WEKA SOFTWARE

WEKA (**Waikato Environment for Knowledge Analysis)** is open-source software specially designed for data mining application to support deep learning algorithms. This software consists of variety of tools which handle numerous properties for data preparation, classification, regression, clustering, association rules mining and visualization in short WEKA is a collection of different visualization tools and algorithms for the analysis of deep learning algorithm and predictive modeling for a given database.

### 4.2 SUPPORT VECTOR MACHINE (SVM)

Support vector classifier is a supervised learning technique best used for linear system classification. This algorithm is used for the analysis of database classification and regression techniques. This algorithm constructs a hyper plane or a set of hyper planes in the multi-dimensional space thereby linearly distinguishing the classes. We can treat the classification to be the best when the distance from the hyper plane to the nearest training set data point is large.[9]

### 4.3 NEURAL NETWORKS

Artificial neural network is the interconnection of neurons which produce output upon the variation of characteristics of input and the activation. Each input is mapped to its unique weight, so to modify the output we alter the weights and the activation function. This process of modification of output based on input and activation is called as the learning rate.

There are different components present in ANN namely:

- Neurons

Each neuron (j) obtains a signal i.e. an input $p_j(t)$ from the preceding neurons, where t is in discrete time parameter. The preceding set of neurons has the following components:

1. An activation $a_j(t)$, depending on the neuron state.
2. Threshold value $\theta$ which remains unchanged until the learning rate is altered.
3. An activation function f for computation of the new state i.e. (t+1) form $a_j(t)$ activation and $p_j(t)$ input state thereby giving us the relation:

$$a_j(t+1) = f(a_j(t), pj(t), \theta)$$

4. Lastly, output function $o_j(t)$ is given by

$$o_j(t)=f_{out}(a_j(t))$$

- Weights and bias

Each neuron is mapped to its unique weights denoted as" wij" where the subscripts I refers to the preceding neuron and j refers to the input neuron. By the alteration of the weights we can modify our outputs.

Bias term is added to the total sum of input weights to serve as thresholds, leading to the shift in the activation function.

- Propagation function

The propagation function is basically used to compute the error rate between the output and the input thereby helping us to get the desired output. This function is given by,

$$P_j(t)=\sum o_j(t)w_{ij} + w_{oj}$$

$w_{ij}$= weight of the input neuron

$w_{oj}$= bias term

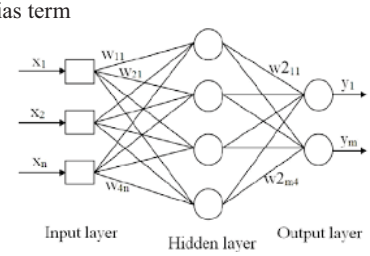


Fig 9 Schematic of Neural network

In this study, MLPNN (multilayer perceptron neural network) is considered which is a supervised training algorithm that can be used for non-linear dataset. MLPNN is a feed-forward artificial neural network consisting of 3-layer nodes namely an input layer, hidden layer and an output layer. The input node uses a nonlinear activation function based on which the back propagation /feed forward ANN algorithm works. Here the learning of the training dataset is based on the connection of the perceptron by changing the connection weights after the processing of which dataset which is compared to the output to find out the error rate [6][10].

## 5. RESULTS AND DISCUSSION

Based on our comparative analysis we found out that, neural networks are prominent in the classification of ECG samples as they have higher accuracy in the detection of abnormality of the signal.

To increase the number of samples we did a comparative analysis on 2 different sets imported MITBIH – arrythmia samples namely:

(1) MIT-BIH atrial fibrillation dataset

(2) ECG-ID Dataset

The number of wave form are 48 and 90 samples respectively. We performed 10-fold cross -validation wherein we allot 1 sample as validation set and rest 9 subsamples as training samples and also testing on 30 records of unknown samples using MLPNN algorithm and SVM machine learning algorithms in Weka platform.

Based on our experimental analysis we got an accuracy of 94% using MLPNN classifier and since our ECG dataset is a non-linear signal, we can conclude that SVM classifier is not suitable for accurate analysis of the signal.

Table 3: CROSS VALIDATION ACCURACY

| **Classifier** | **Accuracy** | **Correctly classifies samples** |
|---|---|---|
| SVM | 85% | 117 |
| MLPNN | 94.5% | 130 |

Table 4 : USING TEST SAMPLES

| **Classifier** | **Accuracy** | **Correctly classifies samples** |
|---|---|---|
| SVM | 72% | 22 |
| MLPNN | 96% | 29 |

## 6. ACKNOWLEDGMENT

We would sincerely thank to the experts who have contributed in development of MIT-BIH database.

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